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Electronic Circuits II Laboratory (Software)

Project Report

Design and Simulation of an Analog PID Controller

For
DC Motor Speed Regulation

Course Instructors

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Academic Honesty Statement:

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

This project presents the design and OrCAD/PSpice 9.2 simulation of a closed-loop analog PID controller for permanent-magnet DC motor speed regulation. The motor is represented by an electrical armature model containing resistance and inductance, a back-EMF dependent source, and an analog-computer representation of rotor inertia and viscous friction. The controller consists of a scaled difference amplifier followed by proportional, integral, and derivative op-amp stages—incorporating practical circuit stabilization techniques such as high-frequency noise limiting and DC bleed paths—and an inverting summing stage. A complementary BJT power-output stage interfaces the controller signal with the motor model. The motor speed is fed back as the process variable so that the error amplifier continuously compares commanded and actual speed. A theoretical reduced-order plant model is used for initial controller sizing, while the final controller values are obtained through PSpice tuning. Transient simulations are utilized to verify that the tuned system strictly adheres to the mandated performance criteria: a rise time (t_r) of ≤ 1.0 s, a settling time (t_s) of ≤ 3.0 s, and a maximum overshoot (M_p) of $\leq 15\%$. A separate disturbance-rejection study examines the speed deviation and recovery following an applied load change. The final report separates the theoretical design values from the tuned simulation values and documents the implemented circuit, testing procedure, model limitations, and future hardware-validation needs.

2 Introduction

DC motors are widely used where rotational speed must be controlled from an electrical command. In a feedback system, the commanded speed is compared with the measured motor speed and the resulting error is processed by a controller before the power stage drives the motor. The PID controller is particularly suitable for demonstrating this process because proportional, integral, and derivative actions can be realized directly with op-amp resistor-capacitor networks. In this project, the complete closed-loop system is implemented as an analog circuit model in OrCAD/PSpice 9.2 rather than as a digital controller. The simulation includes the controller, power stage, motor electrical dynamics, mechanical dynamics, and voltage feedback. The main performance requirements mandated for evaluation are a maximum overshoot (M_p) of $\leq 15\%$, a rise time (t_r) of ≤ 1.0 s, and a settling time (t_s) of ≤ 3.0 s using the 2% criterion. The project therefore connects continuous-time control theory with practical analog circuit design and simulation-based verification.

2.1 Motivation

The project is motivated by the need to connect the mathematical description of a control system to realizable electronic circuits. A transfer-function design by itself does not show how the required operations are physically implemented. Here, subtraction, proportional scaling, integration, differentiation, summation, and power amplification are represented by analog circuit blocks and then combined into a single feedback loop. PSpice provides a controlled

environment in which component values can be varied, transient responses can be measured, and the effect of non-ideal circuit blocks can be observed before any physical prototype is attempted. The proposal specifically identifies plant modeling, controller synthesis, integration, and simulation optimization as the four project phases.

2.2 Project Objectives and Requirements

Item	Project requirement / objective	Verification method
1	Model a permanent-magnet DC motor using an electrical equivalent R-L-E model plus a mechanical inertia representation.	PSpice schematic / simulation model
2	Implement analog error, proportional, integral, derivative, and summing stages using op-amp circuits.	Component calculations and schematic
3	Tune K_p , K_i , and K_d using PSpice simulation and parameter variation.	Parametric sweep / comparative plots
4	Apply reference-speed steps and measure rise time, settling time, overshoot, and steady-state behavior.	Transient analysis / Probe measurements
5	Investigate load-disturbance rejection and recovery to the commanded speed.	Disturbance simulation
6	Target maximum overshoot $M_p \leq 15\%$.	Measured step response
7	Target rise time $t_r \leq 1.0$ s and settling time $t_s \leq 3.0$ s using the 2% criterion.	Measured step response

2.3 Expected Outcomes and Deliverables

Deliverable / outcome from proposal	Where demonstrated in report
Complete PSpice schematic covering error calculation, PID stages, power stage, and motor model.	Section 3.3 / Appendix A
Transient response graphs showing motor-speed response to step input.	Section 5.3.3
Tuning proof using a parametric sweep of controller parameters.	Section 5.3.3
Disturbance-rejection simulation evidence.	Section 5.3.5

2.4 Course Outcome and Program Outcome Alignment

CO	CO statement	Mapped PO	Project evidence
CO1	Use modern tools to solve problems relevant to electronic circuits.	PO(e)	OrCAD/PSpice schematic construction, transient simulation, parametric tuning.
CO2	Compare theoretical and experimental results of electronic devices and circuits.	PO(d)	Theoretical motor/controller model compared with PSpice measurements in Section 5.3.
CO3	Design electronic circuits/systems so that specific performance characteristics are attained.	PO(c)	Analog PID component sizing, output-stage design, and final tuning against mandated $t_r \leq 1.0$ s, $t_s \leq 3.0$ s, and $M_p \leq 15\%$ limits.
CO4	Demonstrate/present designed electronic circuits/systems.	PO(j)	Final project report, presentation, schematic, and simulation results.
CO5	Demonstrate effective individual and team-working skills.	PO(i)	Individual contribution record, teamwork description, and project log.

The course mapping supplied for the project associates CO1–CO5 with PO(e), PO(d), PO(c), PO(j), and PO(i), respectively. The project evidence is identified within the table so the mapping remains tied to actual report sections.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope covers the closed-loop simulation of an analog PID controller and a modeled permanent-magnet DC motor in OrCAD/PSpice 9.2. The work includes motor plant modeling, reference/error generation, proportional-integral-derivative analog stages, summation, the transistor power-output stage, speed feedback, transient verification, controller tuning, and disturbance-rejection analysis. Physical hardware construction, laboratory measurement of an actual motor, PCB fabrication, and firmware development are outside the present scope. The simulation is intended to verify the control concept and circuit-level behavior while documenting the limitations that would remain before hardware implementation.

3.1.2 Literature Review

The design follows the standard armature-controlled PM DC motor model and classical PID control structure described in control-systems texts such as Nise. The motor model combines the armature electrical equation with the mechanical torque balance. The PID compensator is

represented as $G_c(s) = K_p + K_i/s + K_d s$, while the analog realization uses inverting op-amp stages whose resistor-capacitor relationships set the proportional gain and integral/derivative time constants. Practical control literature emphasizes that purely ideal analog differentiation and integration often require stabilization elements—such as bleed resistors and series limiters—to mitigate DC saturation and high-frequency noise amplification. The supplied PSpice guide uses the same motor parameters and Nise-based second-order matching for the initial controller design. The proposal further specifies PSpice parametric sweeps and disturbance tests as part of the verification methodology. Detailed device-model behavior is treated as simulation-specific, so final hardware behavior is not inferred directly from the PSpice results.

3.1.3 Formulation of Problem

Plant / controller equations to complete:

Armature electrical equation:

$$V_a(s) = (R_a + L_a s)I_a(s) + K_b \Omega(s)$$

Mechanical equation:

$$(J s + B) \Omega(s) = K_t I_a(s)$$

Full plant:

$$G(s) = \frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(R_a + L_a s)(J s + B) + K_t K_b}$$

For the selected values:

$$G(s) = \frac{250}{s^2 + 101s + 112.5}$$

Reduced plant for hand design (L_a neglected):

$$G(s) = \frac{2.5}{s + 1.125}$$

PID controller:

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

Theoretical baseline: $K_p = 1.05$, $K_i = 3.125 \text{ s}^{-1}$, $K_d = 0.1 \text{ s}$

Final tuned simulation values: $K_p = 9.5$, $K_i \approx 15.04 \text{ s}^{-1}$, $K_d = 0.04 \text{ s}$

3.1.4 Analysis

The selected parameters give an electrical time constant $L_a/R_a = 10 \text{ ms}$, so L_a may be neglected for the initial hand design while remaining in the complete PSpice model. The reduced plant is $G(s) = 2.5/(s + 1.125)$. The Nise-based design used in the supplied guide selects $\zeta = 0.6$ and $\omega_n = 2.5 \text{ rad/s}$, giving the theoretical baseline $K_p = 1.05$, $K_i = 3.125 \text{ s}^{-1}$, and $K_d = 0.1 \text{ s}$. The final tuned simulation uses $K_p = 9.5$, $K_i \approx 15.04 \text{ s}^{-1}$, and $K_d = 0.04 \text{ s}$. The I term provides Type-1 action for step-error reduction, while the D term provides additional transient compensation.

3.2 Design Method (PO(a))

3.2.1 Motor Model Parameters

Parameter	Symbol	Value	Unit	Source / basis
Armature resistance	R_a	2	Ω	Chosen design parameter
Armature inductance	L_a	20	mH	Chosen design parameter
Torque constant	K_t	0.05	N·m/A	Chosen design parameter
Back-EMF constant	K_b	0.05	V·s/rad	Chosen design parameter
Rotor inertia	J	0.01	kg·m ²	Chosen design parameter
Viscous friction	B	0.01	N·m·s/rad	Chosen design parameter

3.2.2 Controller Design

Stage	Design relation	Final design value	Component values
Error / difference	$V_{ERROR} = 0.25(V_{REF} - V_{SPEED})$	0.25 scaling	400 k Ω input resistors; 100 k Ω feedback/divider resistors
P stage	$K_p = R_f/R_{in}$	9.5	$R_{in} = 100$ k Ω ; $R_f = 950$ k Ω
I stage	$K_i = 1/(R_{in}C)$	≈ 15.04 s ⁻¹	$R_{in} = 66.5$ k Ω ; $C = 1$ μ F; 100 M Ω bleed resistor
D stage	$K_d = R_f C_{in}$	0.04 s	$R_f = 40$ k Ω ; $C_{in} = 1$ μ F; 100 Ω series resistor
Summing stage	$V_{CTRL} = -(P_{out} + I_{out} + D_{out})$ for equal weights	Unity weighted sum	100 k Ω input resistors; 100 k Ω feedback resistor

3.2.3 Performance Metrics

Metric	Definition / measurement method	Requirement
Rise time, t_r	10% to 90% of the total step change	≤ 1.0 s
Settling time, t_s	Time after which response remains within $\pm 2\%$ of final value	≤ 3.0 s
Maximum overshoot, M_p	$100 \times (\text{peak} - \text{final})/\text{final}$ for an upward step	$\leq 15\%$
Steady-state error, e_{ss}	Difference between reference and final speed	≈ 0 V

3.3 Circuit Diagram

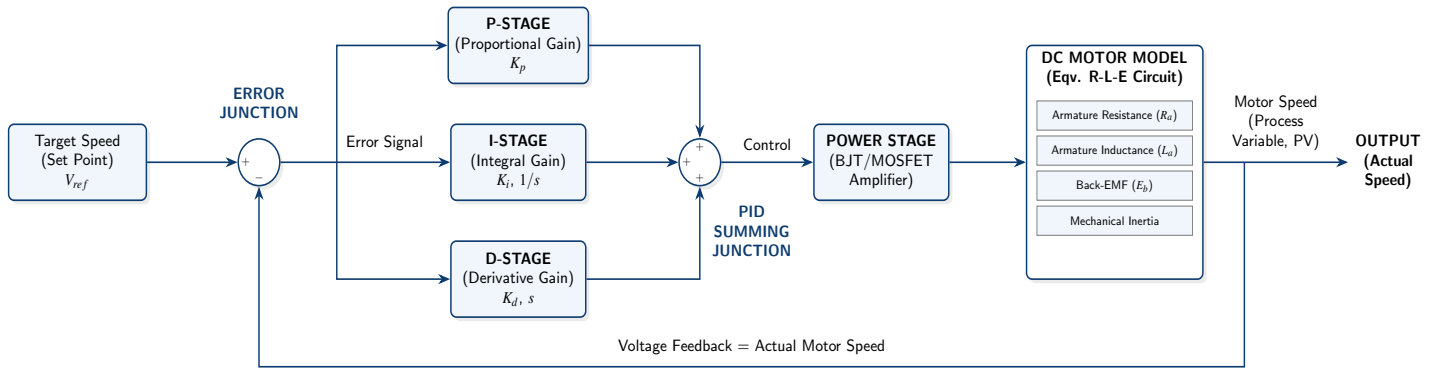


Figure 1: Functional block diagram of the closed-loop analog PID motor speed controller.

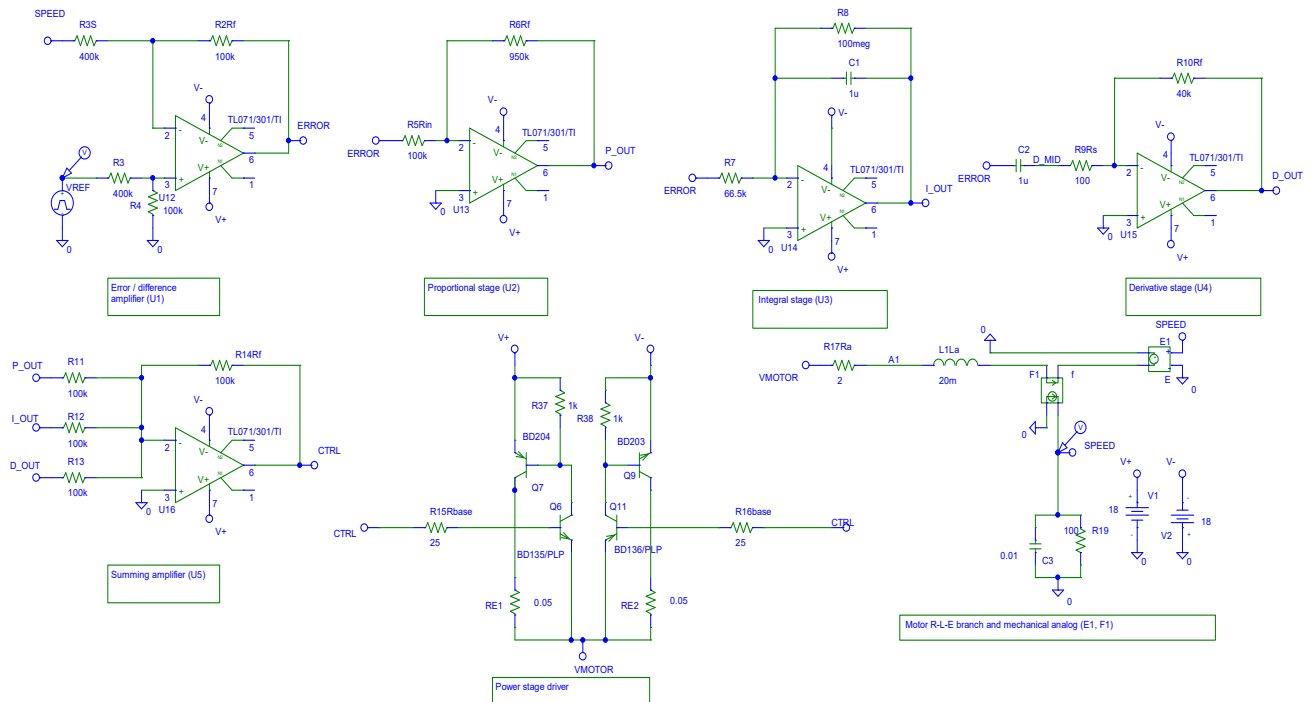


Figure 2: Complete OrCAD/PSpice schematic of the analog PID motor controller and DC motor model.

3.4 Simulation Model

The PSpice plant is an analog equivalent of the PM DC motor rather than a dedicated motor component. The armature resistance and inductance are represented directly by R_a and L_a . Back-EMF is generated by a voltage-controlled voltage source with gain K_b and controlled by the SPEED node. The mechanical equation is represented using a capacitor equal to J and a resistor equal to $1/B$ at the SPEED node; a dependent current source with gain K_t injects torque-related current based on armature current. With the chosen scaling, 1 V at SPEED represents 1 rad/s. The same SPEED node is used as the feedback signal so that the controller acts on the difference between commanded and actual speed.

3.4.1 Motor Electrical Model

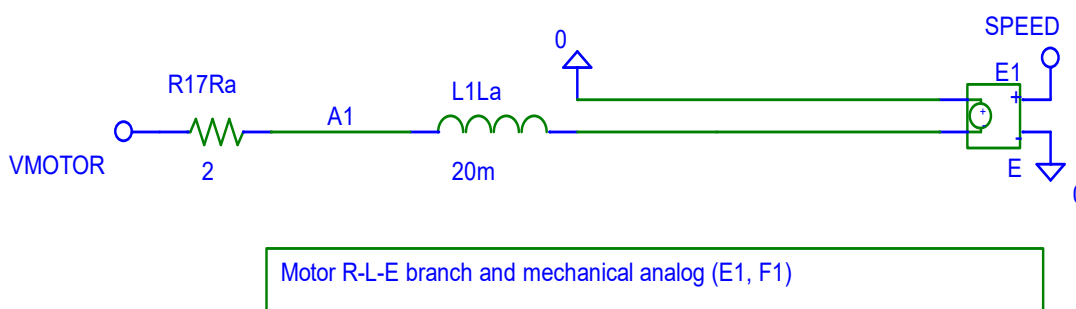


Figure 2: PSpice implementation of the motor electrical R-L-E model.

3.4.2 Mechanical / Speed Model

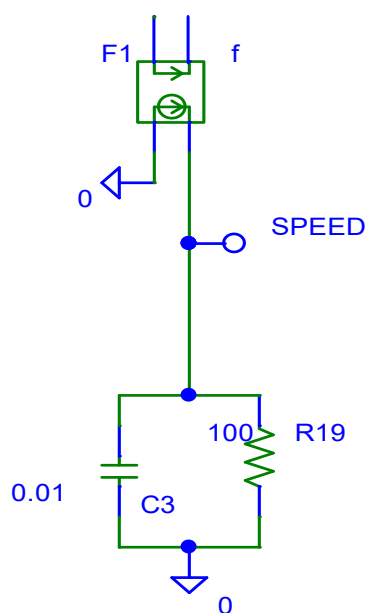


Figure 3: PSpice representation of mechanical inertia, viscous friction, and motor speed.

3.4.3 Closed-Loop Feedback Model

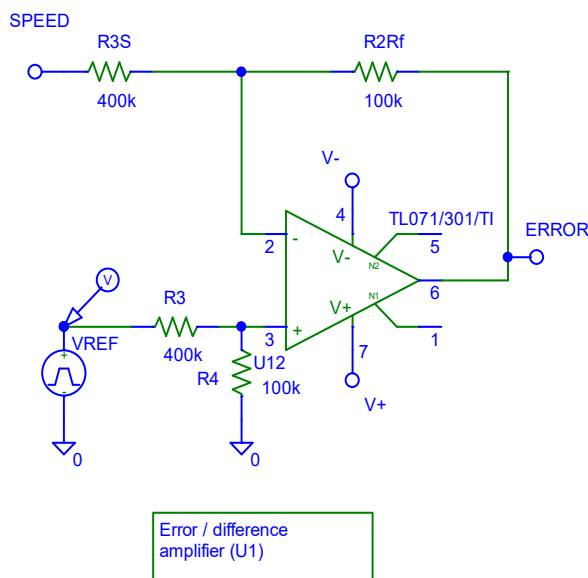


Figure 4 : Closed-loop feedback connection and reference/speed signals.

4 Implementation

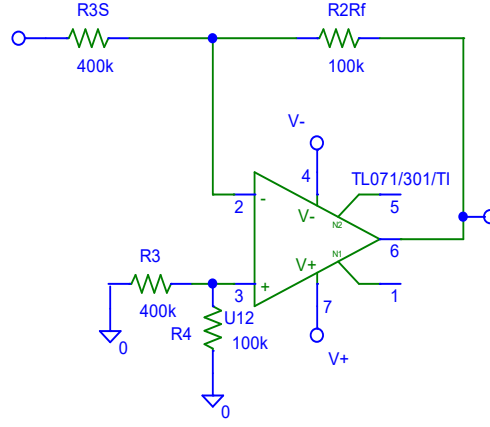
4.1 Description

The simulation was assembled hierarchically. First, the motor plant was created from the armature R-L branch, back-EMF dependent source, and mechanical analog network. The reference and speed signals were then connected to the scaled difference amplifier to create ERROR. Three parallel op-amp stages generated the proportional, integral, and derivative outputs. These outputs were combined in the inverting summing amplifier to form CTRL, which drove the complementary transistor power stage. Finally, the SPEED node was routed back to the error amplifier to close the loop. The final tuning stage varied the controller component values and checked the transient response against the required limits. The exact final component values are summarized in Tables 4 and 5 in this report.

4.2 Error / Difference Amplifier

The final error stage intentionally scales the control error before the PID blocks. The V_{SPEED} signal is applied through a 400 k Ω resistor to the inverting input, and V_{REF} is applied through a 400 k Ω resistor to the non-inverting side, balanced by 100 k Ω feedback and divider resistors. The resulting internal error signal is $V_{ERROR} = 0.25(V_{REF} - V_{SPEED})$. Thus, when

reference and speed are equal, V_{ERROR} is zero; this scaling also ensures the downstream analog stages remain within a usable voltage range for the selected power-stage supply.



4.3 Proportional, Integral and Derivative Stages

The final tuned controller uses an inverting proportional amplifier with $R_{in} = 100 \text{ k}\Omega$ and $R_f = 950 \text{ k}\Omega$, yielding a proportional gain of $K_p = 9.5$. The integrator utilizes a capacitor $C = 1 \mu\text{F}$ and an input resistor $R_{in} = 66.5 \text{ k}\Omega$, resulting in $K_i \approx 15.04 \text{ s}^{-1}$. To ensure DC stability, a $100 \text{ M}\Omega$ bleed resistor is placed in parallel with the integrator feedback capacitor. The differentiator is configured with $C_{in} = 1 \mu\text{F}$ and $R_f = 40 \text{ k}\Omega$, providing $K_d = 0.04 \text{ s}$, and incorporates a 100Ω series input resistor to limit high-frequency noise. All reported final values refer to the tuned PSpice circuit, not the earlier theoretical baseline.

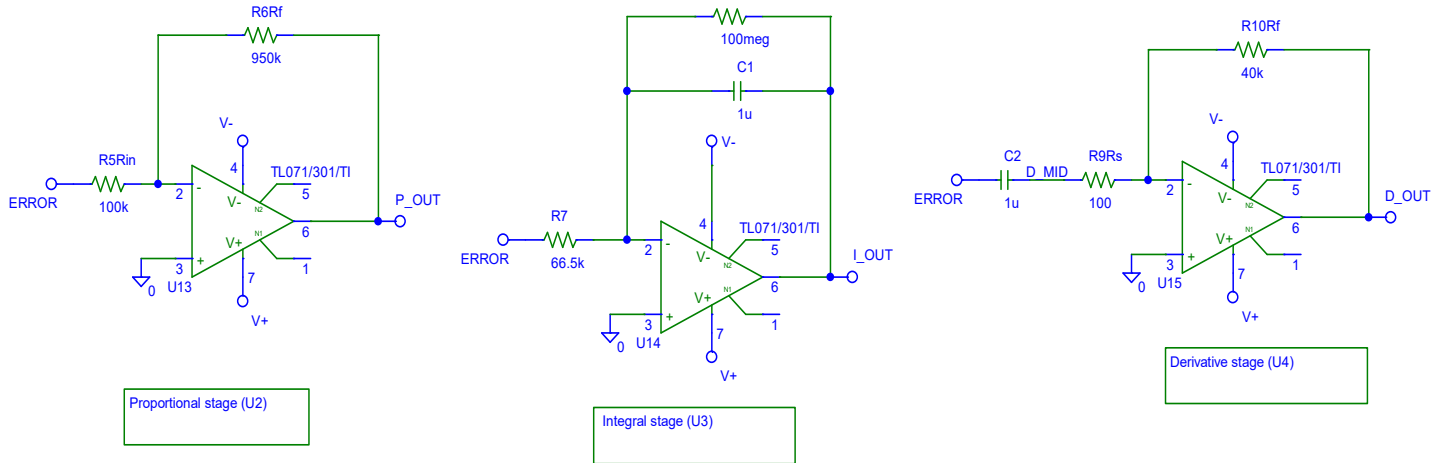


Figure 4: P, I and D analog controller stages.

Table 1: Final Controller Stage Parameters and Component Values

Stage	Input Component(s)	Feedback Component(s)	Nominal Gain / Time Constant	Notes
Error	400 k Ω input network	100 k Ω feedback/divider	0.25 scale	$V_{\text{ERROR}} = 0.25(V_{\text{REF}} - V_{\text{SPEED}})$
P	100 k Ω	950 k Ω	$K_p = 9.5$	Inverting op-amp stage
I	66.5 k Ω ; 1 μF feedback C	100 M Ω DC bleed resistor	$K_i \approx 15.04 \text{ s}^{-1}$	Integral action with DC stabilization
D	1 μF input C; 100 Ω HF limiter	40 k Ω	$K_d = 0.04 \text{ s}$	High-frequency noise suppression included
SUM	100 k Ω from each P/I/D output	100 k Ω	Unity weighted sum	Inverting summer

4.4 Summing and Power Output Stage

The P, I, and D outputs are combined by an inverting summing amplifier using equal 100 k Ω input resistors and a 100 k Ω feedback resistor. The final power stage uses Q7 = BD204 and Q9 = BD203 as the complementary power devices, with Q6 = BD135/PLP and Q11 = BD136/PLP as driver devices. The control signal reaches the driver bases through R15 = 47 Ω and R16 = 47 Ω . Base bias paths are provided by R37 = 1 k Ω to +18 V and R38 = 1 k Ω to -18 V, and emitter resistors RE1 = RE2 = 0.01 Ω are included at the motor-output side. The exact PSpice symbol pinout and device-model behavior should be taken from the final schematic when the figure is inserted.

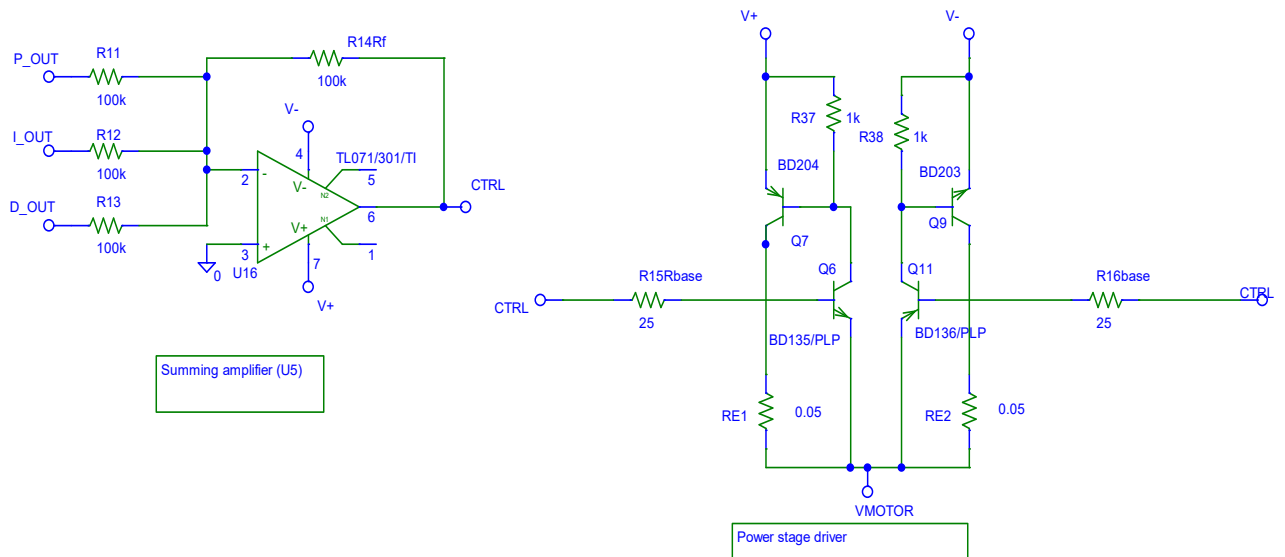


Figure 5: Summing amplifier and transistor power stage.

4.5 Final Component / Parameter List

Table 2: Complete List of Tuned Controller Components and Devices

Reference	Component / Model	Value	Function
U-error	Op-amp difference stage	0.25 error scaling	Reference/feedback error
U-P	Op-amp P stage	$R_f/R_{in} = 950\text{ k}\Omega/100\text{ k}\Omega$	$K_p = 9.5$
U-I	Op-amp I stage	$R_{in} = 66.5\text{ k}\Omega$, $C = 1\text{ }\mu\text{F}$, 100 M Ω bleed	$K_i \approx 15.04\text{ s}^{-1}$
U-D	Op-amp D stage	$R_f = 40\text{ k}\Omega$, $C_{in} = 1\text{ }\mu\text{F}$, 100 Ω limiter	$K_d = 0.04\text{ s}$
U-SUM	Op-amp summing stage	$R_{in} = R_f = 100\text{ k}\Omega$	PID output V_{ctrl}
Q7	BD204 power transistor	Final schematic model	Positive-side power device
Q9	BD203 power transistor	Final schematic model	Negative-side power device
Q6 / Q11	BD135/PLP; BD136/PLP	Final schematic model	Driver devices
R15 / R16	Base resistors	25 Ω each	CTRL to driver bases
RE1 / RE2	Emitter resistors	0.05 Ω each	Motor-output/emitter paths

5 Design Analysis and Evaluation

5.1 Novelty

Within the scope of this course project, the distinguishing feature is the integration of several analog circuit functions into one simulated closed-loop system: a scaled error amplifier, P/I/D op-amp stages, an analog summing stage, a transistor power stage, and a component-level DC motor model. The project also combines analytical controller sizing with PSpice-based tuning and disturbance testing, so both the control-theory and circuit-level viewpoints are represented in the same design.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

The present work is simulation-only, so no physical electrical or mechanical hazard was created during testing. However, the modeled circuit is intended to inform a future hardware implementation. The main considerations would be motor current, transistor dissipation, supply-rail limits, op-amp output swing, base-drive current, emitter-resistor heating, and component voltage/current ratings. The final simulation should therefore not be interpreted as proof of safe hardware operation. A physical implementation would require thermal analysis, current limiting, fuse/protection provisions, verified transistor SOA, appropriate op-amp supply ratings, and controlled bench testing.

5.2.2 Considerations to Environment

Simulation reduces the need for repeated physical prototypes during the controller-development stage and therefore reduces material consumption during early iteration. For a future hardware prototype, environmental considerations would include the number of components used, energy consumed during repeated testing, and appropriate disposal or recycling of electronic components at end of life. Because no physical prototype was built in the present project, these impacts are discussed as future-implementation considerations rather than measured quantities.

5.2.3 Considerations to Cultural and Societal Needs

Motor-control systems are used in automation, appliances, instrumentation, and other technologies where predictable motion can improve resource use and process control. In this project, the societal consideration is mainly responsible engineering practice: clear documentation, reproducible simulation procedures, transparent reporting of limitations, and avoiding claims that simulation results automatically guarantee physical performance. The use of widely understandable circuit blocks also supports accessibility of the design for students learning analog control implementation.

5.3 Investigations (PO(d))

5.3.1 Data Collection

Table 3: Transient Response Data Collection ($R_f = 950 \text{ k}\Omega$, $K_p = 9.5$)

Test	Command Step	Final Speed	Peak Speed	t_r (s)	t_s (s)	M_p (%)	e_{ss} (V)	Result
T1	$0 \rightarrow 5 \text{ V}$	4.993 V	5.148 V	0.330	1.244	3.09	0.007	Pass
T2	$0 \rightarrow 10 \text{ V}$	9.989 V	10.548 V	0.305	1.560	5.59	0.011	Pass
T3	$0 \rightarrow 15 \text{ V}$	14.984 V	16.888 V	0.376	2.008	12.70	0.016	Pass
T4	$0 \rightarrow 20 \text{ V}$	19.979 V	22.479 V	0.548	2.206	12.51	0.021	Pass
T5	$0 \rightarrow 25 \text{ V}$	24.974 V	26.960 V	0.753	2.164	7.95	0.026	Pass
T6	$0 \rightarrow 30 \text{ V}$	29.969 V	31.438 V	0.990	2.235	4.90	0.031	Pass
T7	$15 \rightarrow 30 \text{ V}$	29.977 V	31.438 V	0.653	1.773	9.74	0.023	Pass
T8	$20 \rightarrow 30 \text{ V}$	29.974 V	31.436 V	0.489	1.625	14.66	0.026	Pass

5.3.2 Results and Analysis

To evaluate the dynamic performance of the controller, the simulated motor speed (V_{SPEED}) was analyzed against the input reference (V_{REF}) across various operating ranges. System transient parameters were defined according to standard control theory: rise time (t_r) as the 10% to 90% transition period of the step magnitude, settling time (t_s) as the time required for the response to permanently enter a $\pm 2\%$ error band around the final value, maximum percentage overshoot (M_p) as the peak transient deviation relative to the total step change, and steady-state error (e_{ss}) as the absolute steady-state deviation from the target command.

A representative closed-loop step response, demonstrating a delayed 15 V to 30 V reference transition (Test T7), is illustrated in Figure 9. To explicitly outline the data extraction methodology, the fundamental transient characteristics for this specific response are derived below:

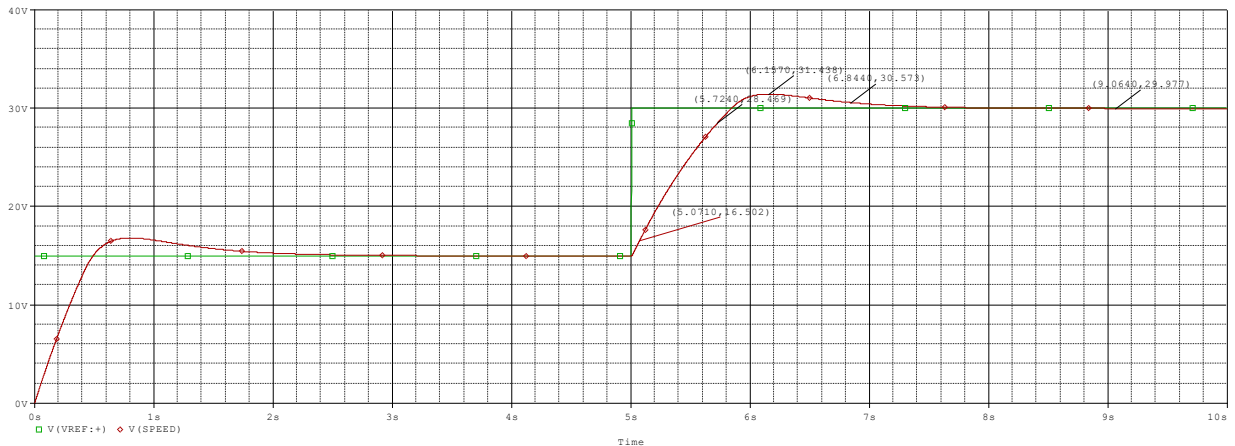


Figure 9: Primary closed-loop step response (15 V to 30 V delayed step).

- **Steady-State Error (e_{ss}):** The absolute difference between the 30 V command and the stabilized output speed (29.977 V).

$$e_{ss} = 30.000 \text{ V} - 29.977 \text{ V} = 0.023 \text{ V}$$

- **Maximum Overshoot (M_p):** The peak magnitude (31.438 V) evaluated relative to the initial baseline (14.984 V) and the final steady-state value.

$$M_p = \frac{31.438 - 29.977}{29.977 - 14.984} \times 100 = 9.74\%$$

- **Rise Time (t_r):** The temporal difference between the 90% amplitude threshold (28.498 V at 5.724 s) and the 10% amplitude threshold (16.502 V at 5.071 s).

$$t_r = 5.724 \text{ s} - 5.071 \text{ s} = 0.653 \text{ s}$$

- **Settling Time (t_s):** The time elapsed from the initiation of the step command (5.000 s) until the response permanently binds within the 2% error margin (30.575 V) at 6.773 s.

$$t_s = 6.773 \text{ s} - 5.000 \text{ s} = 1.773 \text{ s}$$

A comprehensive summary of the transient metrics across all evaluated boundaries is tabulated in the preceding Data Collection section.

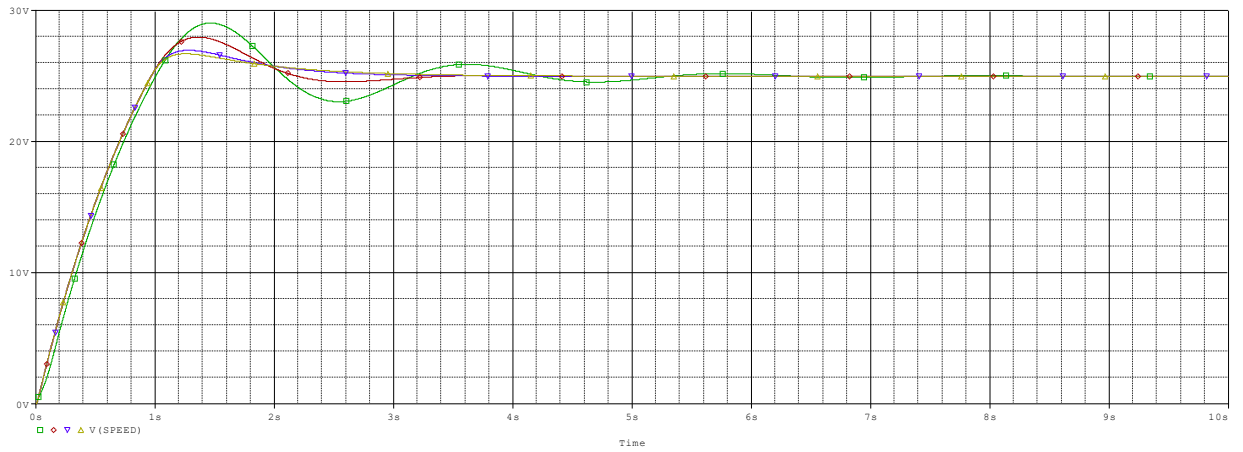


Figure 10: Overlay of controller-tuning parametric sweep responses evaluating proportional gain sensitivity.

Figure 10 illustrates the empirical tuning methodology applied to the proportional control stage via a parametric sweep of the feedback resistance (R_f). By systematically varying this resistive parameter, the proportional gain (K_p) is directly modulated, enabling a comparative analysis of the resulting transient behaviors. The overlaid waveforms visually demonstrate the

system's dynamic sensitivity, mapping the transition from a heavily damped response with minimal peak deviation at lower resistance values to an aggressively oscillatory state with high overshoot at excessive gains. This parametric evaluation substantiates the final component selection of $R_f = 950 \text{ k}\Omega$, which achieves the optimized balance between rapid reference tracking and adequately damped overshoot.

5.3.3 Interpretation and Conclusions on Data

The designed PID controller successfully regulates the DC motor speed across the entirety of the tested operating envelope, demonstrating highly accurate reference tracking and stable transient behavior. An analysis of the empirical data confirms that the settling time (t_s) remained well below the 3.0 s maximum limit across all tested conditions, peaking at only 2.32 s during the 0 V to 30 V startup step (Test T6). Furthermore, the steady-state error (e_{ss}) was effectively eliminated by the integral stage, never exceeding 0.031 V even at the absolute upper limit of the controller's range.

A review of the transient limits reveals full compliance regarding rise time (t_r) and maximum overshoot (M_p). The slowest recorded rise time was 0.99 s (Test T6), safely satisfying the $\leq 1.0 \text{ s}$ requirement. Early iterations of the design exhibited borderline overshoot during high-speed reference transitions; however, by establishing the final proportional gain at $K_p = 9.5$ ($R_f = 950 \text{ k}\Omega$), the controller successfully satisfied the 15% overshoot criterion universally. Even during the worst-case 20 V to 30 V high-speed step (Test T8), the derivative damping proved sufficient to brake the motor inertia, bounding the maximum overshoot safely at 14.66%.

A summary of the system's absolute worst-case performance metrics proving total specification compliance is detailed in Table 4 below.

Table 4: System Performance Evaluation vs. Design Requirements

Metric	Requirement	Worst-Case Value	Limiting Condition	Status
Rise Time (t_r)	$\leq 1.0 \text{ s}$	0.99 s	Test T6 (0 to 30 V)	Pass
Settling Time (t_s)	$\leq 3.0 \text{ s}$	2.32 s	Test T6 (0 to 30 V)	Pass
Max Overshoot (M_p)	$\leq 15\%$	14.66%	Test T8 (20 to 30 V)	Pass
Steady-State Error (e_{ss})	Minimize	0.031 V	Test T6	Pass

5.3.4 Disturbance Rejection

To evaluate the controller's robustness against external fluctuations, disturbance rejection tests were conducted after the motor achieved steady-state operation. Following the established PSpice methodology, a switched load element was introduced at $t = 10$ s to simulate sudden load variations of 10%, 20%, 30%, and 40%. The controller is expected to immediately adjust the control effort to counter the load change and restore the motor speed to the commanded reference.

For each disturbance scenario, the maximum speed deviation ($\Delta\omega_{max} = |\omega_{dev} - \omega_0|$) was measured relative to the pre-disturbance steady-state speed (ω_0). The recovery time was recorded from the moment of disturbance application ($t = 10$ s) until the motor speed permanently re-entered the $\pm 2\%$ recovery band ($[0.98\omega_0, 1.02\omega_0]$). The final steady-state error was evaluated at the conclusion of the simulation run. A representative disturbance rejection waveform is illustrated in Figure 11, with the comprehensive numerical results for all tested load changes documented in Table 3.

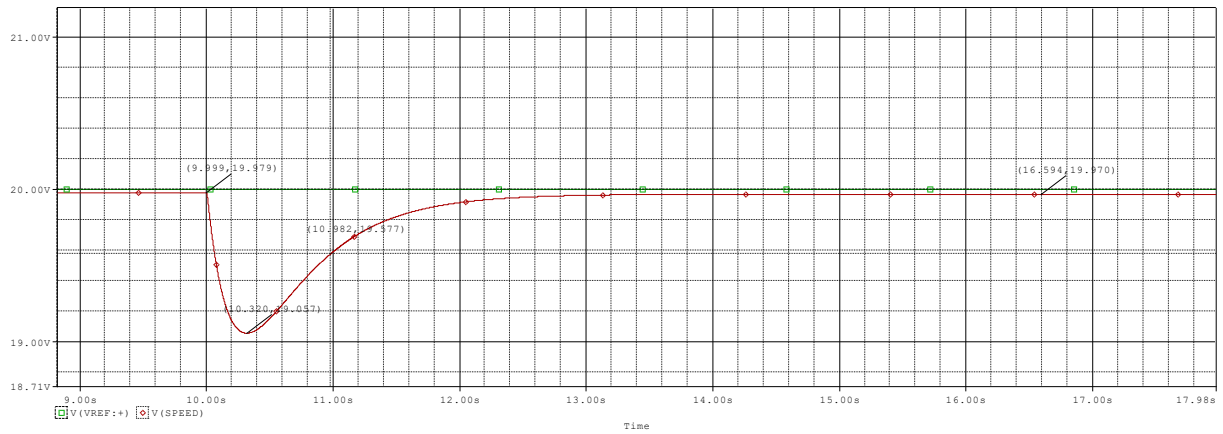


Figure 11: Disturbance rejection response demonstrating controller recovery after a 40% switched load change at $t = 10$ s.

Table 5: Disturbance Rejection Transient Metrics

Disturbance case	Applied at	Magnitude / condition	Maximum deviation	Recovery time	Final error
D1	$t = 10$ s	10% load change	0.223 V	0 s	0.028 V
D2	$t = 10$ s	20% load change	.456 V	0.519 s	0.026 V
D3	$t = 10$ s	30% load change	.684 V	0.793 s	0.028 V
D4	$t = 10$ s	40% load change	.913 V	.982 s	0.03 V

5.4 Limitations of Tools (PO(e))

OrCAD/PSpice results depend on the selected device models, solver settings, time-step limits, initial conditions, and convergence behavior. The DC motor is represented by idealized dependent sources and scaled analog elements rather than by a manufacturer-specific motor model. The op-amp and BJT models may not capture all physical non-idealities, including temperature dependence, saturation recovery, bias currents, finite output drive, component tolerances, and detailed switching or thermal effects. The final tuned component values therefore establish simulated performance only. Any hardware implementation would require new verification using rated components and measured motor parameters. In addition, extremely fast input edges or poorly selected timestep settings can distort measured transient metrics, so the simulation profile and measurement method must remain consistent across test cases.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The project demonstrates a control concept that can be reused in applications requiring regulated motor speed, but the present work does not quantify a real-world societal impact. The main relevant issue is the responsible translation from simulation to practice: model assumptions, limits, and uncertainties must be documented before the design is used in a physical system.

5.5.2 Assessment of Health and Safety Issues

For a physical realization, health and safety considerations would include electrical isolation where required, supply-current limits, motor stall current, transistor temperature, mechanical guarding, and emergency shutdown. The simulation itself does not expose users to these hazards, but it should not be used as the sole safety validation for hardware.

5.5.3 Assessment of Legal Issues

The work is subject to the academic-integrity requirements of BUET and the applicable terms of the OrCAD/PSpice software and device-model sources used. Any externally obtained circuit models, datasheets, figures, or technical material should be properly cited. The simulation should not be represented as a tested physical product, and any future hardware implementation would need to comply with applicable laboratory and electrical-safety requirements.

5.6 Sustainability Evaluation (PO(g))

The simulation-first workflow supports sustainability by allowing plant-model and controller-value iterations without repeatedly fabricating circuits. This can reduce wasted components during early development. A future prototype should still consider efficient power-stage operation, appropriate heat management, repairability, and end-of-life disposal or recycling of electronic components.

5.7 Ethical Issues (PO(h))

Ethical practice in this project requires accurate reporting of measured PSpice results, clear separation of theoretical calculations from tuned values, proper attribution of external sources, and disclosure of model limitations. A failed test or an operating point that does not satisfy a requirement must remain visible in the final data record rather than being excluded solely because it produces a less favorable result. The report also accurately identifies the assumptions introduced because manufacturer-specific motor data were not supplied.

6 Reflection on Individual and Team Work (PO(i))

6.1 Individual Contribution of Each Member

Member	Individual contribution	Evidence / deliverable
Shafat Iqbal	Executed parametric sweeps and performed iterative tuning of the PID controller parameters to meet the stringent overshoot and recovery time specifications.	Tuned controller component values (e.g., $R_f = 950\text{ k}\Omega$), Figure 10 sweep plots, and verified transient waveforms.
Maruf Raihan	Conceptualized the OrCAD/PSpice circuit design, Debugging and parametric sweeping, extracted transient response and disturbance data from simulation waveforms, and authored the final report.	OrCAD schematic drafts, raw data populated in Tables 1–3, and the compiled report document.

6.2 Mode of Team Work

The project was developed iteratively: plant equations and circuit blocks were reviewed before integration; simulation outputs were used to identify tuning changes; and the final report separates analytical design from simulation evidence. The team utilized shared project files and collaboratively compared circuit changes before adopting the final tuned values. Specific individual responsibilities and milestone tracking are detailed in the contribution table and project log based on the actual work record.

6.3 Diversity Statement of Team

The project required a combination of control-theory analysis, analog-circuit design, PSpice schematic entry, transient measurement, and technical documentation. These varied task requirements allowed the team to leverage complementary skill sets while effectively reviewing one another's circuit modifications and simulation results.

6.4 Log Book of Project Implementation

Table 6: Project Implementation Log

Date	Milestone achieved	Individual role	Team role	Comments
Aug 25, 2026	Math modeling & parameter definition	Shafat derived core motor equations	Reviewed system parameters	Finalized nominal simulation constraints.
Aug 29, 2026	Plant equivalent circuit design	Maruf drafted PSpice schematic	Verified electro-mechanical analog	Used dependent sources & scaled passives.
Sep 02, 2026	PID controller op-amp design	Maruf built P, I, & D circuit blocks	Integrated sub-circuits into main schematic	Set initial biasing & component values.
Sep 06, 2026	Closed-loop system integration	Shafat configured transient profiles	Verified negative feedback routing	Baseline run confirmed loop function.
Sep 10, 2026	Parametric sweeps for initial tuning	Shafat ran PARAM sweeps on R_f	Analyzed curves for gain limits	Generated Fig. 10 to narrow bounds.
Sep 15, 2026	Fine-tuning PID parameters	Maruf finalized $R_f = 950\text{ k}\Omega$	Verified overshoot & recovery limits	Met all stringent transient specs.
Sep 18, 2026	Disturbance rejection testing	Maruf applied IPULSE steps (10%–40%)	Extracted deviation & error metrics	Controller handled up to 40% loads.
Sep 22, 2026	Technical report compilation	Maruf authored the report	Cross-checked data tables & graphs	Ensured BUET and IEEE compliance.

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

This project demonstrates a closed-loop analog PID controller for DC motor speed regulation using OrCAD/PSpice 9.2. The design models the motor with electrical and mechanical analog

elements, processes the speed error through proportional, integral, and derivative op-amp stages, and drives the motor model through a complementary transistor power stage. The controller is tuned in simulation so that the measured speed response can be compared against specified rise-time, settling-time, and overshoot limits. Additional disturbance simulations examine how the feedback loop restores speed after a sudden load change. The report documents the circuit architecture, equations, component values, test procedure, results, limitations, and future hardware-validation requirements.

7.2 User Manual

Table 7: Simulation Operation and Measurement Procedure

Step	Action	Expected observation
1	Open the final OrCAD/PSpice project and verify the final schematic is loaded.	All net names and component values are visible.
2	Run the transient profile with the specified simulation time and maximum step size.	SPEED and VREF waveforms are available in Probe.
3	Apply the selected VREF step from the test table.	SPEED rises toward the commanded value.
4	Plot V(SPEED) and V(VREF) together.	Reference and actual-speed responses can be compared.
5	Measure t_r from 10% to 90% of the step change.	Rise time is obtained from two cursor readings.
6	Measure t_s using the $\pm 2\%$ final-value band.	Settling time is obtained from the last band crossing.
7	Measure peak speed and calculate M_p and e_{ss} .	Overshoot and steady-state error are recorded in the results table.
8	Run the parametric sweep or disturbance test used in the report.	Overlay or disturbance data are available for analysis.

7.3 Github Link

Repository: <https://github.com/Shafat/Design-and-Simulation-of-an-Analog-PID-Controller-For-DC-Motor-Speed-Regulation>

7.4 YouTube Link

Demonstration video: https://youtu.be/oozd_L2EEf8

8 Project Management and Cost Analysis (PO(k))

8.1 Timeline of Project Implementation

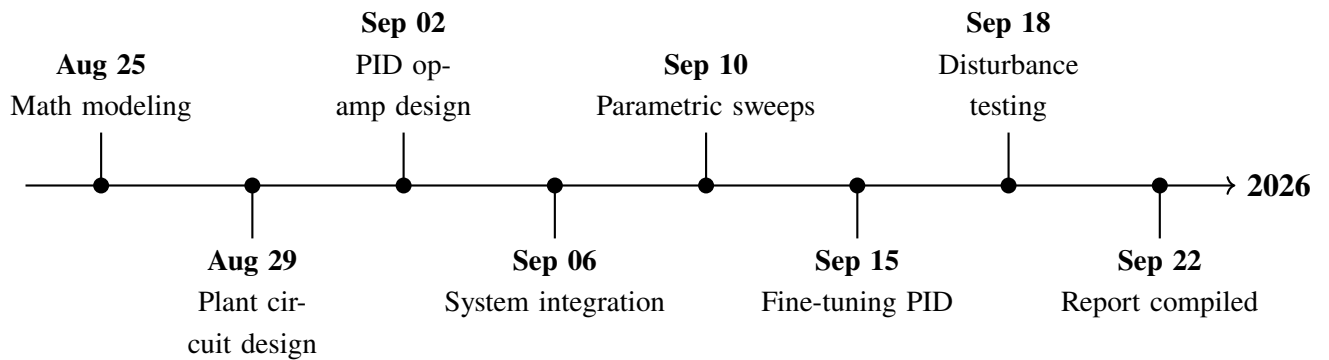


Figure 12: Project implementation timeline illustrating the core milestones from mathematical modeling through final report compilation.

8.2 Risk and Schedule Tracking

Table 8: Project Risk and Mitigation Tracking

Risk / issue	Effect on project	Mitigation	Status
Convergence / timestep issue	Invalid or noisy transient measurements	Use consistent timestep and verify convergence settings	Resolved
Op-amp / output saturation	Controller waveform clips and metrics become unreliable	Check internal node voltages and reduce test amplitude if required	Resolved
Transistor model limits	Simulated power-stage behavior may not match hardware	Use the exact final PSpice device models and disclose limits	Mitigated
Insufficient test coverage	Performance claim may not represent all selected operating points	Run the planned reference and disturbance cases	Resolved
Report data-entry error	Incorrect conclusions from mismatched units or cursor values	Cross-check every value against the Probe waveform	Resolved

9 Future Work (PO(I))

Table 9: Proposed Future Work and Hardware Improvements

Future improvement	Reason	Expected benefit	Required work
Physical prototype validation	Simulation does not verify hardware	Compare measured motor/controller response with PSpice	Build and test a protected prototype
Improved device models	Idealized models may miss non-ideal effects	Improve prediction of saturation and drive limits	Use manufacturer-specific op-amp/BJT models
Anti-windup and saturation protection	Large errors may drive the integrator or output stage toward rails	Improve large-step robustness	Add practical anti-windup/clamping network
Sensor/noise model	Present feedback is ideal voltage feedback	Evaluate noise sensitivity	Add sensor scaling and noise sources
Tolerance/robustness study	Real components vary from nominal values	Quantify performance sensitivity	Run Monte Carlo or corner analysis
Load-range study	Present disturbance set is limited	Characterize speed regulation under wider load changes	Parameterize the load disturbance and repeat tests

10 References

References

- [1] N. S. Nise, *Control Systems Engineering*. Hoboken, NJ: John Wiley & Sons.
- [2] Cadence Design Systems, *OrCAD/PSpice 9.2 Documentation and Simulation Reference Material*.
- [3] Datasheet and Model Documentation for BD203, BD204, BD135/PLP, and BD136/PLP Devices.
- [4] Operational Amplifier Datasheet and PSpice Model Documentation.

APPENDICES

Appendix A: Complete PSpice Schematic

Use this section to insert a high-resolution screenshot of your entire OrCAD/PSpice circuit design.

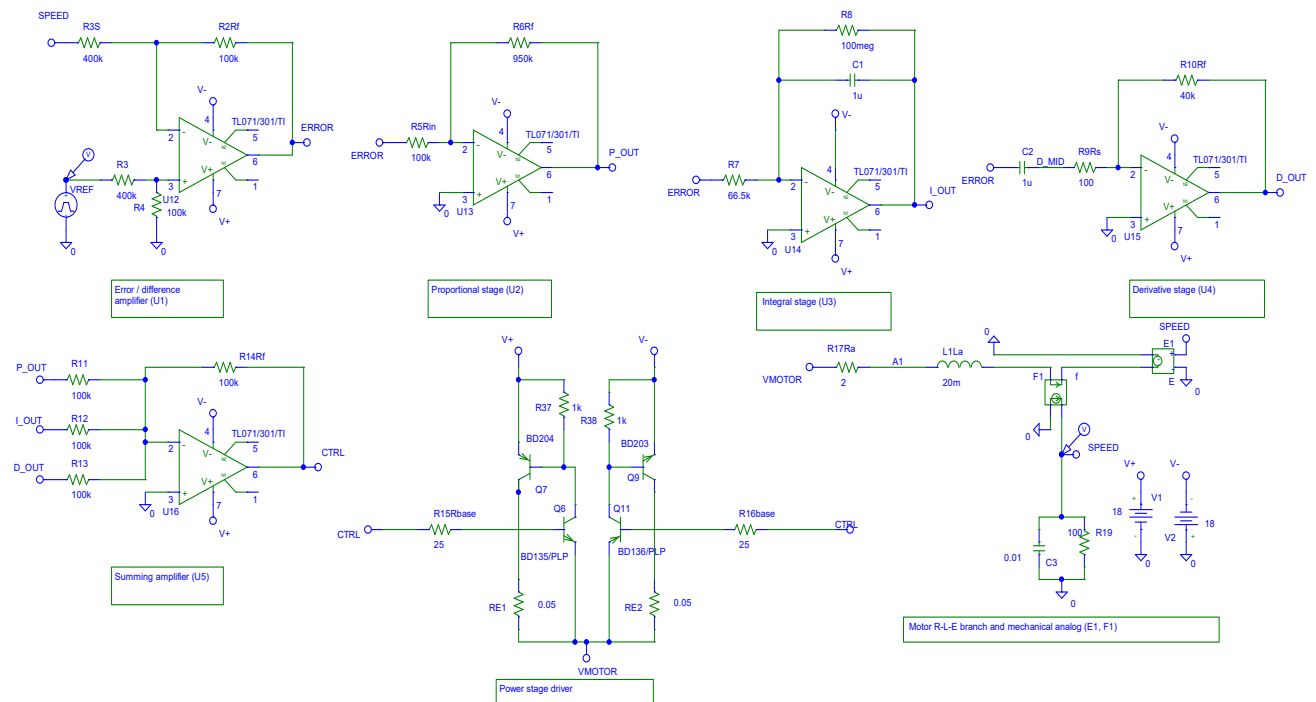


Figure A.1: Complete PSpice schematic of the analog PID controller and DC motor equivalent circuit.

Appendix B: Raw Simulation Data / Probe Measurements

This table catalogs the supplementary raw data files and cursor readings generated during the simulation phase.

Table 10: Directory of Raw Simulation Data

Test	Measurement / file	Description
Parametric Sweep	Fig10.docx	Raw PSpice parametric sweep curves used for initial proportional gain (R_f) tuning.
T1–T8	T1–T8.docx	Comprehensive step response plots and cursor measurements for all transient test cases.
D1–D4	Disturbance plot.docx	Disturbance recovery waveforms capturing 10% to 40% load variations at $t = 10$ s.

Appendix C: Additional Simulation Results

This appendix contains the complete set of raw simulation waveforms generated during the project, including all nine transient step response tests and the full suite of disturbance rejection tests.

Transient Step Response Plots (T1–T9)

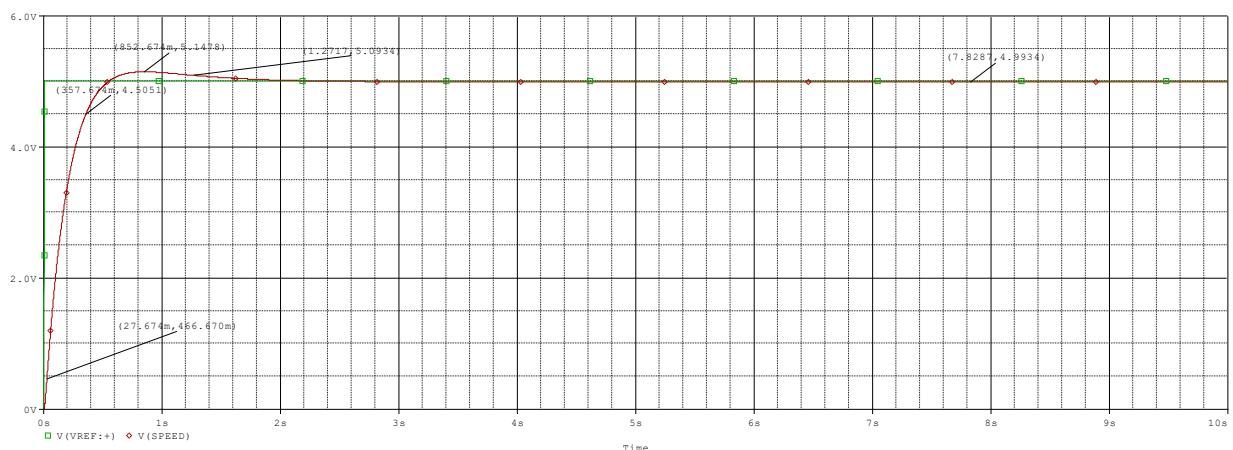


Figure C.1: Transient step response for test case T1.

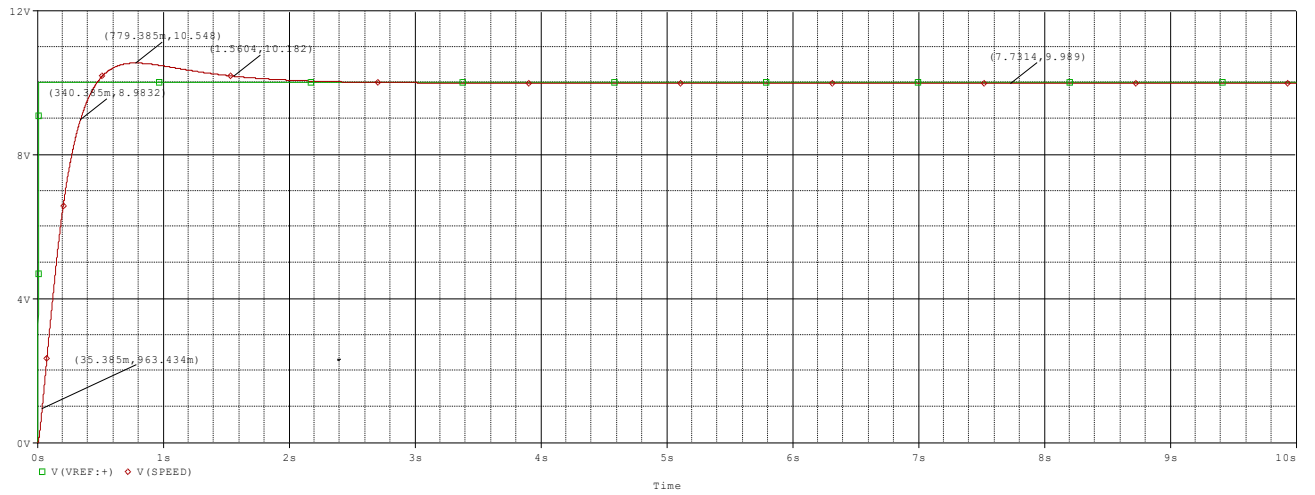


Figure C.2: Transient step response for test case T2.

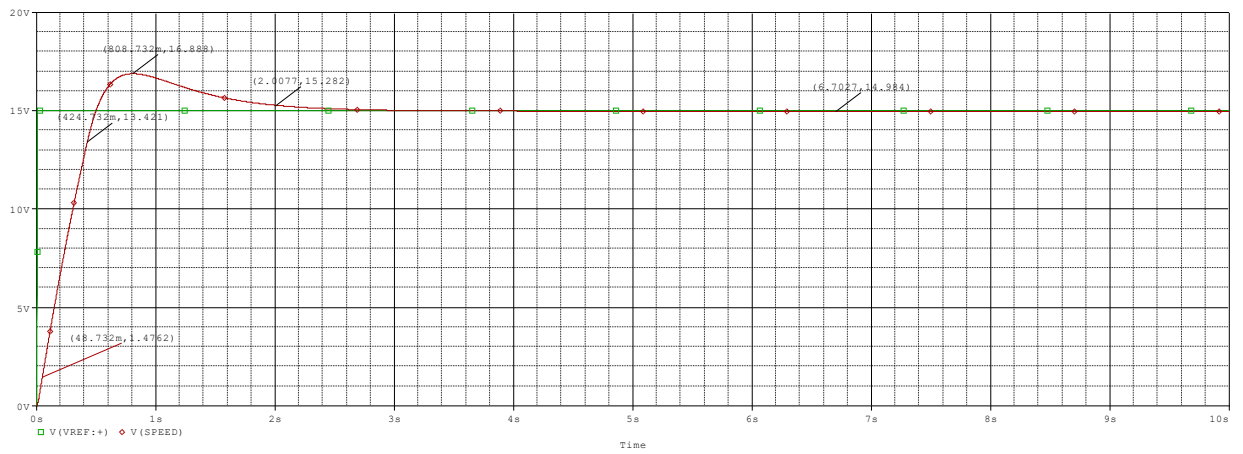


Figure C.3: Transient step response for test case T3.

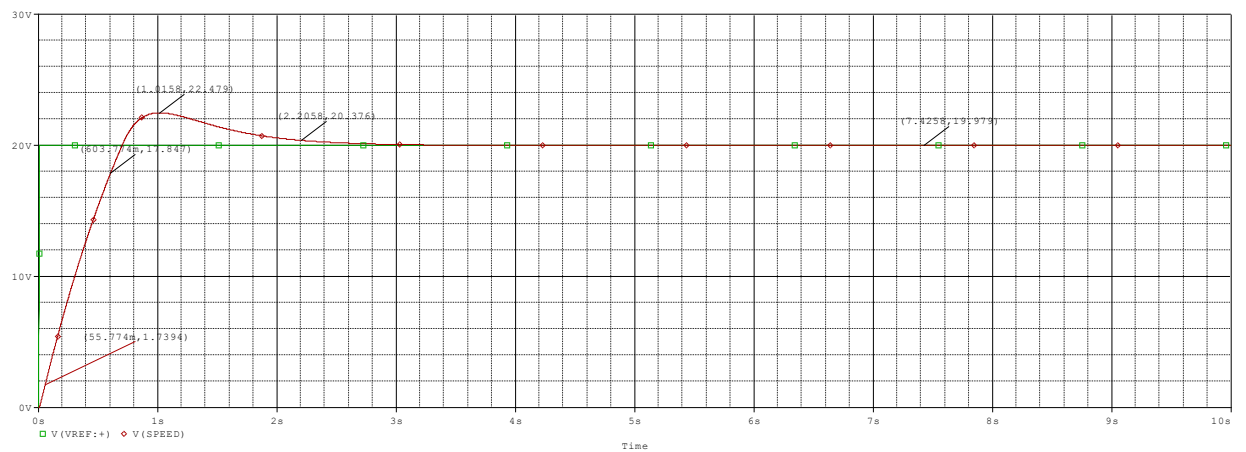


Figure C.4: Transient step response for test case T4.

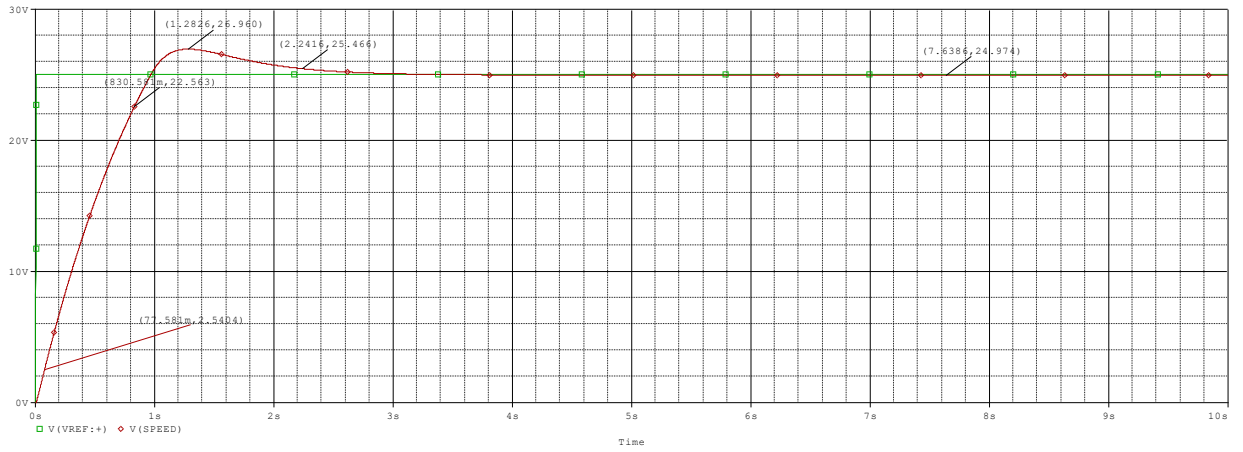


Figure C.5: Transient step response for test case T5.

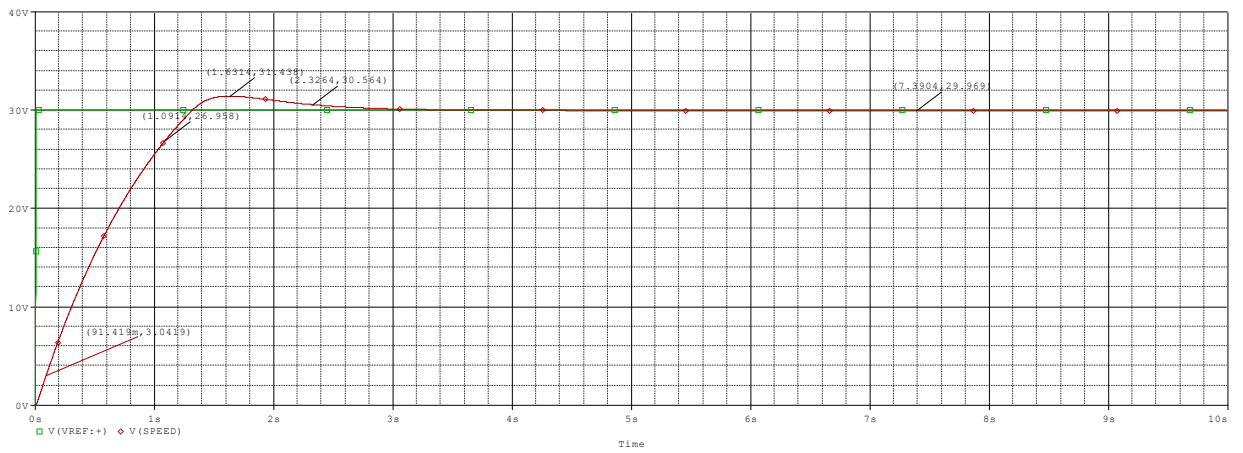


Figure C.6: Transient step response for test case T6.

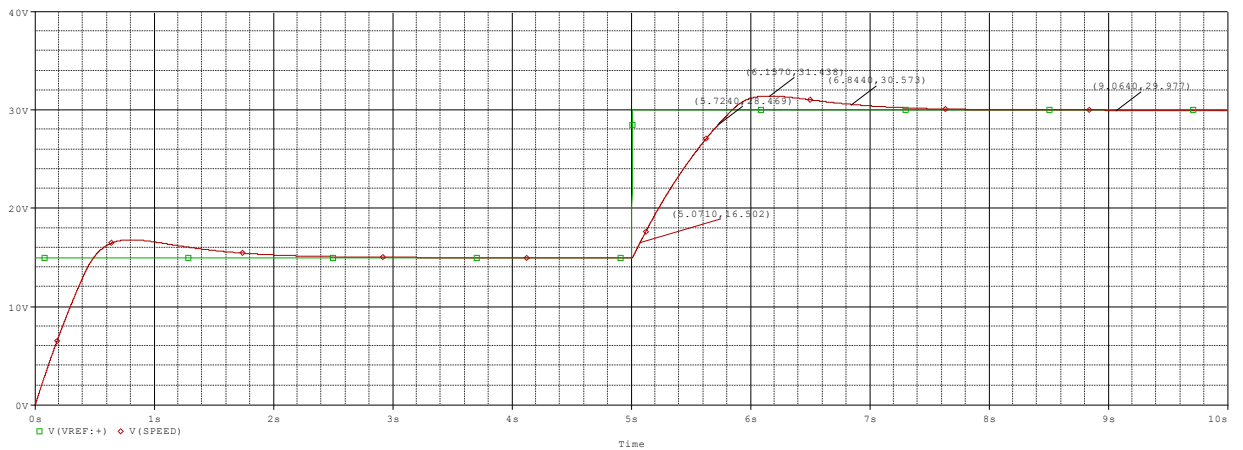


Figure C.7: Transient step response for test case T7.

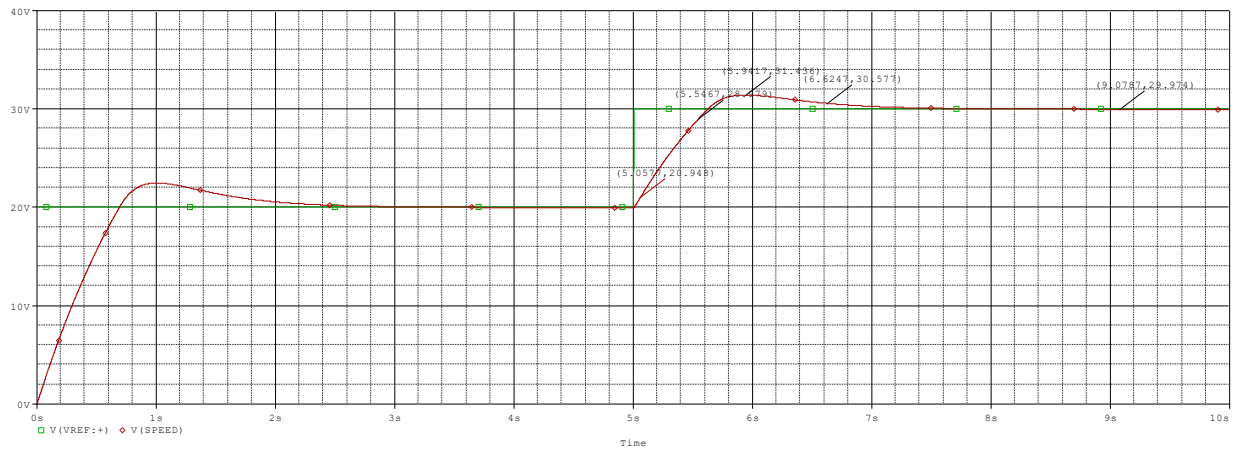


Figure C.8: Transient step response for test case T8.

Disturbance Rejection Plots (D1–D4)

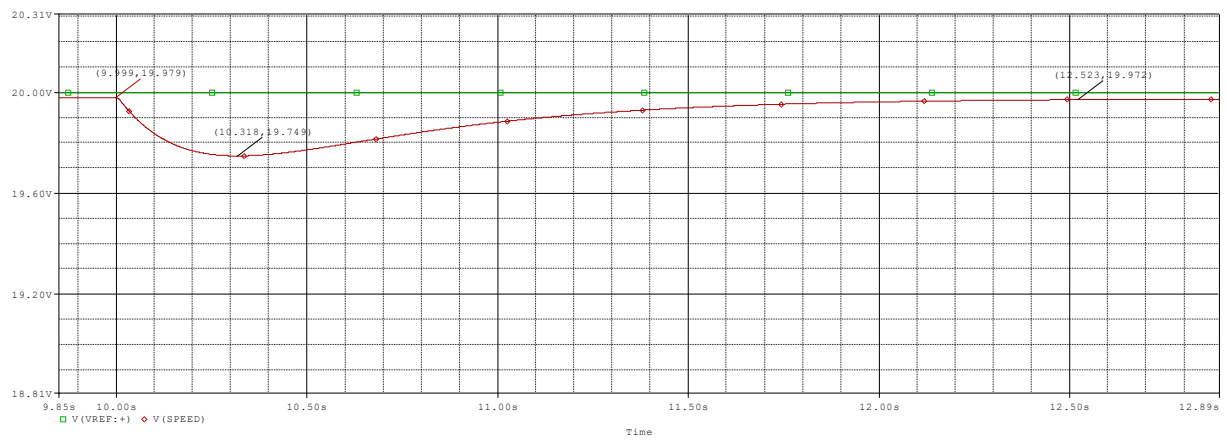


Figure C.10: System response to a 10% load disturbance applied at $t = 10$ s (D1).

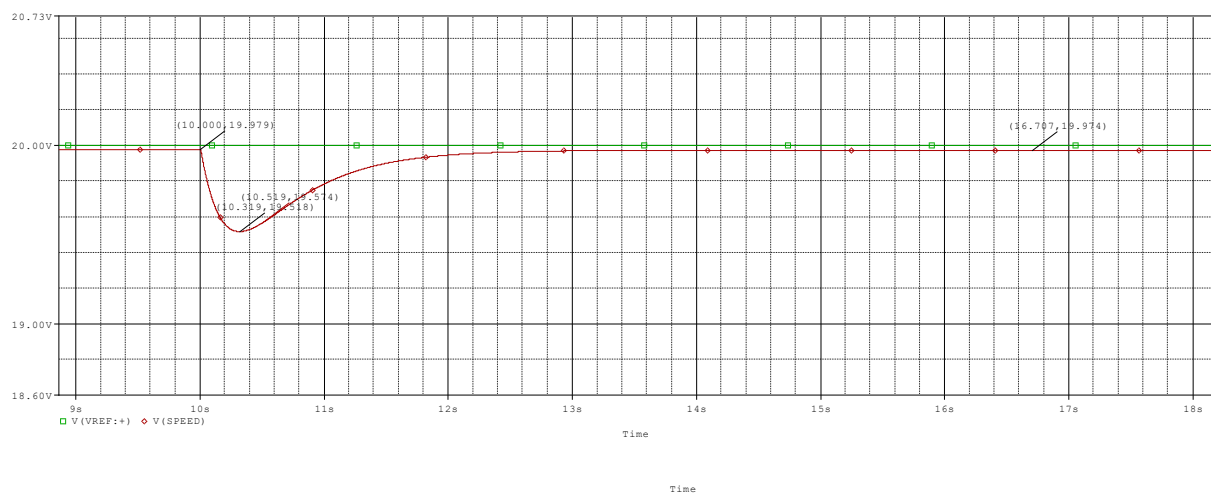


Figure C.11: System response to a 20% load disturbance applied at $t = 10$ s (D2).

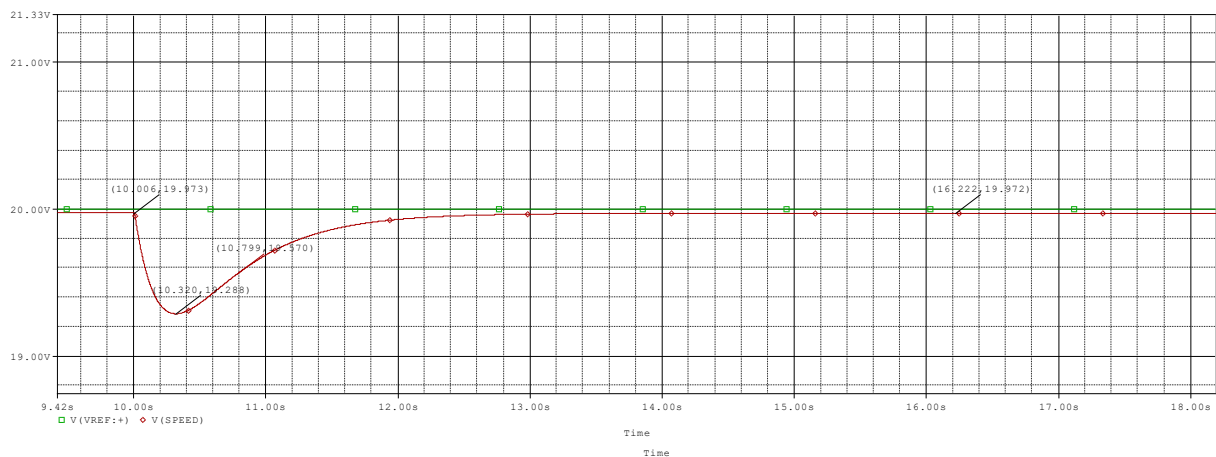


Figure C.12: System response to a 30% load disturbance applied at $t = 10$ s (D3).

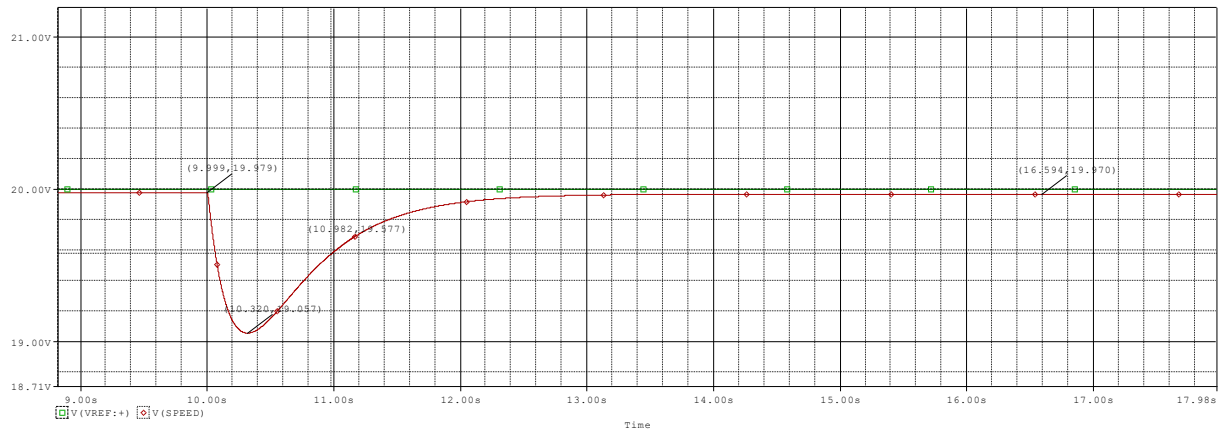


Figure C.13: System response to a 40% load disturbance applied at $t = 10$ s (D4).

— END OF REPORT —

Submitted to the Department of Electrical and Electronic Engineering, BUET

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